Phase 1 report: natural motif lengths in the humpback whale song

Notebook: `Whale_Song_Length_Range_Experiment.ipynb` · Plan: `Length_Range_and_Forward_Search_Plan.md` · Output: `whale_length_ranges.json`

Summary

The experiment measures, for motif lengths from 0.3 s to 25 min, how much more the whale song repeats itself than a random signal with the same spectrum would. It does this for the four levels of humpback song (unit, phrase, theme, song) and for two MFCC coefficients (c0: overall band energy; c1: spectral tilt, the PDF's "coefficient 2").

level	c0 range	c1 range	decided by
unit	1.7–4.2 s	2.7–5.2 s	bump in the detrended correlation (overlaps the 1–3 s prior)
phrase	8.9–120 s (peak 38 s)	8.9–120 s (peak 50 s)	half-maximum width of the excess over the null
theme	75–95 s	60–300 s (peak 95 s)	bump (c0); excess over the null (c1)
song	7–9 min	5–10 min	copies of the motif recur once per song cycle

Main findings:

- **Phrase period ≈ 11 s.** The sharpest regularity in the recording: a few-second chunk's best match is usually 11.3 s later (both coefficients), with autocorrelation peaks at ~10–11 s and its multiples (~20, ~30, ~40 s).
- **Repetition is strongest for ~40–100 s chunks.** Across all bands the excess over the null has one broad maximum there, which corresponds to several repeated phrases (themes).
- **Song cycle ≈ 13 min.** c1's autocorrelation peaks at 788 s (13.1 min), and 5–10 min motifs recur every 11–13 min with regular gaps, matching the PDF's 12–14 min cycle and the 10–20 min prior.
- **c1 is the better coefficient for long motifs.** c0 is dominated by slow loudness changes and shows no excess repetition at song lengths.
- **Alignment.** The 37.5-minute file `SOCAL_H_76_230427_014845` starts exactly **22,424.5 s** (6 h 13 min 44.5 s) into the 10-hour file; the features of the two files agree with r = 0.997 (c0) and 0.996 (c1).

Data and features

- **Recording:** `SOCAL_H_76_230426_193500_df100.x.x.wav`, 10 h at 2 kHz, starting 2023-04-26 19:35:00.
- **Features:** MATLAB-style MFCC at 2 kHz: 12 mel bands from 133 Hz to 1 kHz (MATLAB's default band edges cut at the 1 kHz Nyquist), 0.1 s Hamming window, 10 ms hop, i.e. **100 frames/s** and 3.6 million frames. Columns: log energy, c0, c1, …, c11. Cached in `whale_mfcc_193500_100hz.npz`.
- **37.5-minute file:** the original 200 kHz recording, decimated ×100 to 2 kHz and given the same features (`whale_mfcc_014845_100hz.npz`). It is located inside the 10-hour file by cross-correlating the raw 2 kHz audio at three probe points (all give 22,424.5 s, audio r = 1.000).

Method

Bands

Each level is scanned at a frame rate that keeps motif lengths between ~30 and ~1,500 points. All matrix profiles use an exclusion zone of the full window length, so a window's best match never overlaps it.

level	lengths scanned	rate	data
unit	0.3–8 s, 16 log-spaced	100 Hz	8 × 5-min segments spread over the night

phrase	5–120 s, 12 log-spaced	10 Hz	3 × 1-h segments (start, middle, end)
theme	60–480 s, 10 log-spaced	2 Hz	full 10 h
song	300–1500 s, 1-min steps	1 Hz	full 10 h

Criteria

For each length L, the matrix profile (`stumpy.stump`) gives every window's correlation with its best non-overlapping match, $r = 1 - d^2/(2L)$.

1. **Excess repetition over a null (primary).** *E(L)* = top-decile *r* of the data – top-decile *r* of a *phase-randomized surrogate* (same power spectrum, so the same smoothness and the same bias toward short lengths, but no repeated shapes). A natural motif length shows up as a peak of *E(L)*.
2. **Detrended correlation bumps.** A quadratic trend in log *L* is removed from the best-pair and top-decile *r*; positive bumps are the PDF's "bump" criterion.
3. **Repetition periods.** Peaks of the autocorrelation and of the histogram of offsets between strongly matching windows and their nearest neighbours.
4. **Regularity (song level).** For the best pair at each length, copies are collected as in the PDF (threshold lowered until no gap longer than 20 min remains). A good song length has copies recurring once per song period (± 25%), gap coefficient of variation ≤ 0.3 and no hole longer than 1.5 periods.

Decision rule

E(L) is smoothed over 3 neighbouring lengths, and the range is its half-maximum width around the peak. It is accepted if the peak excess is ≥ 0.02, the peak is **localized** (not at the first or last length of the band and the range does not span the whole band) and a bump or a period supports it. Otherwise:

- song level: the regularity criterion decides;
- other levels: the positive part of the detrended top-decile bump with the largest excess;
- if nothing applies, the initial guess is used (this did not happen).

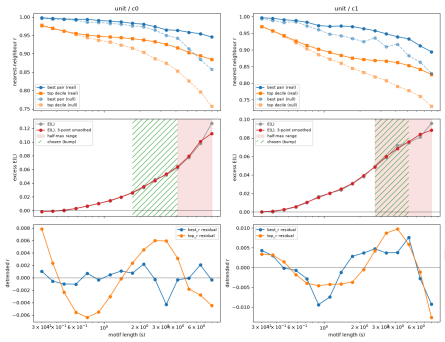
The **threshold** for Phase 2 is the null's top-decile correlation at each length: a detection must be at least as similar as the best 10% of chance matches.

Changes from the plan

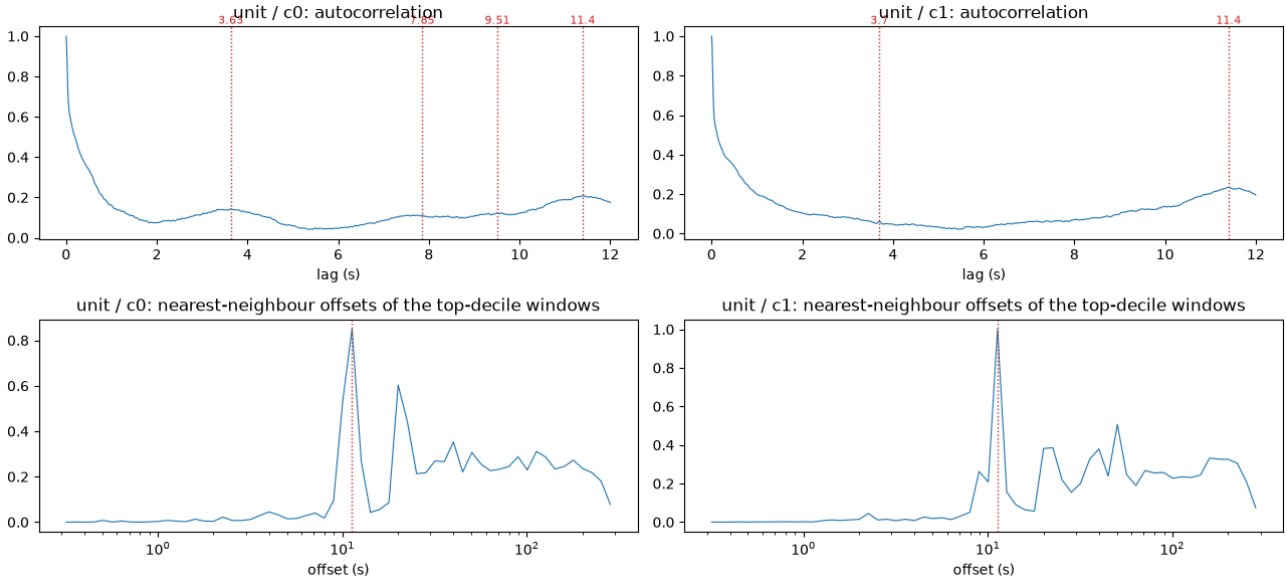
- **Null model.** The planned block-shuffled null was replaced by the phase-randomized surrogate. The jumps where shuffled blocks join made the shuffled c0 series *more* self-similar than the real one (null top-decile r ≈ 0.96).
- **Localized peak required.** In the unit band, *E(L)* keeps rising up to the 8 s edge, so the plain half-maximum rule picked 4–8 s, which is just the edge of the scan. The localization requirement and the bump fallback were added for this case.

Results by level

Units (0.3–8 s)

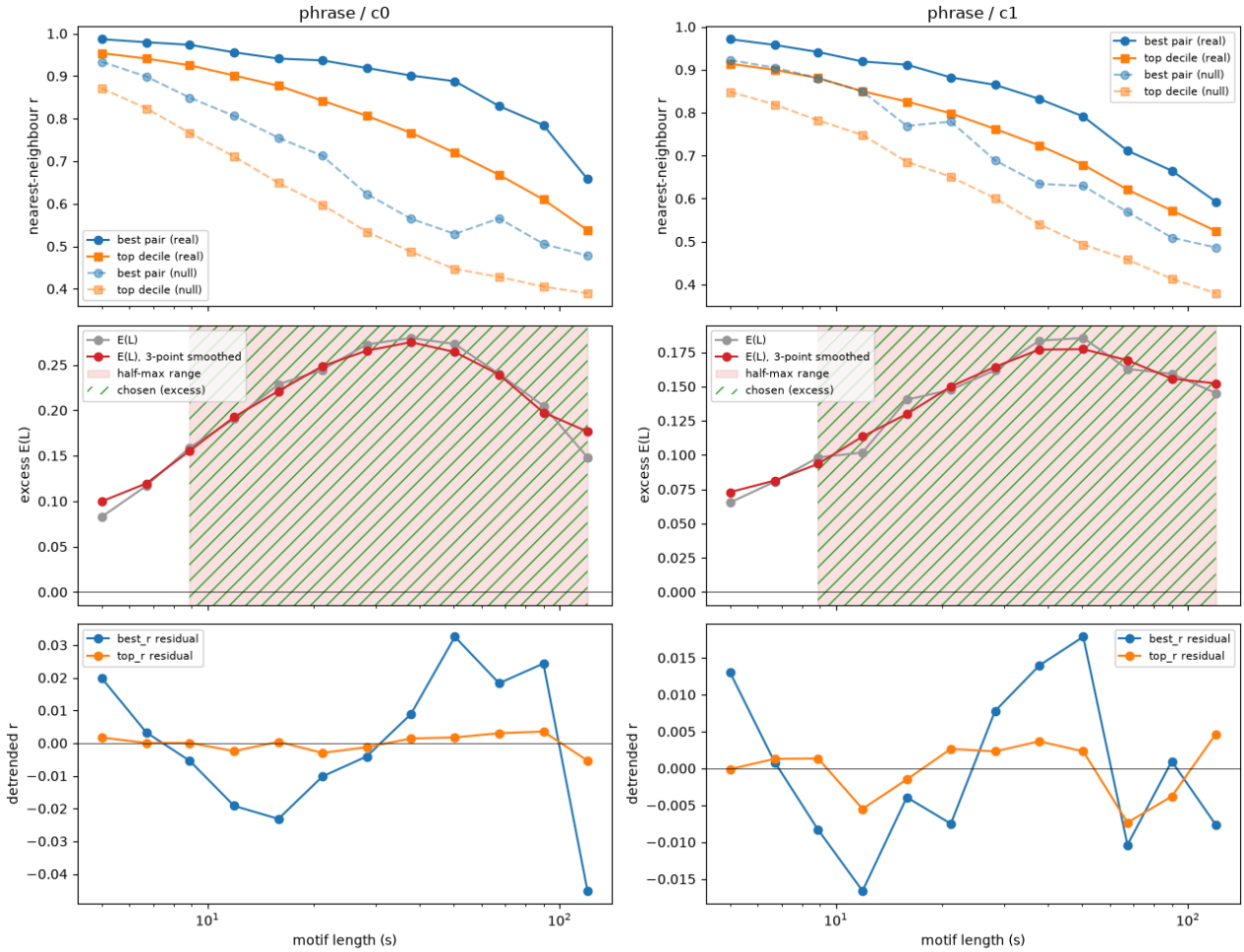


E(L) rises steadily from 0 at 0.3 s to 0.13 (c0) and 0.10 (c1) at 8 s: longer chunks repeat increasingly more than chance, but there is no peak inside the band. The detrended top-decile correlation has a clear bump around **2.7 s (c0)** and **4.2 s (c1)**, and the positive part of that bump gives the unit ranges 1.7–4.2 s and 2.7–5.2 s. Both overlap the 1–3 s prior; a motif window includes the unit plus some of the gap around it.



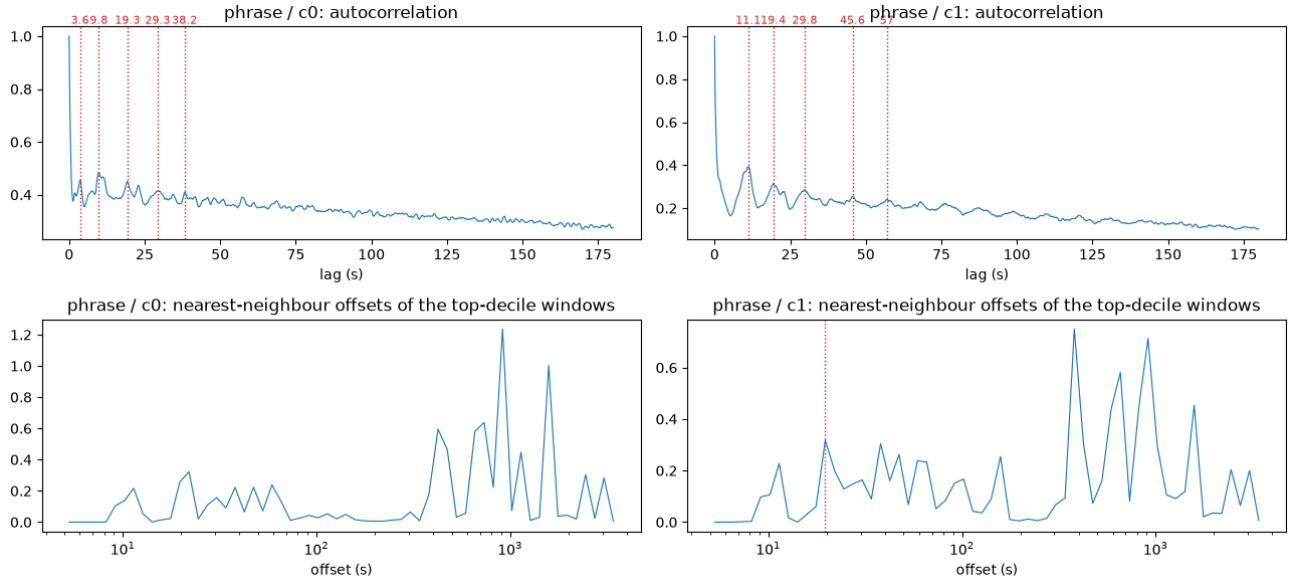
The offset histogram has a very sharp peak at **11.3 s** for both coefficients (and a second one at ~20–22 s): the best match of a strongly repeating few-second chunk is usually one phrase later. The autocorrelation also peaks at 11.4 s, with weaker peaks at 3.6–3.7 s (c0 also 7.9 and 9.5 s), consistent with units repeating within a phrase.

Phrases (5–120 s)



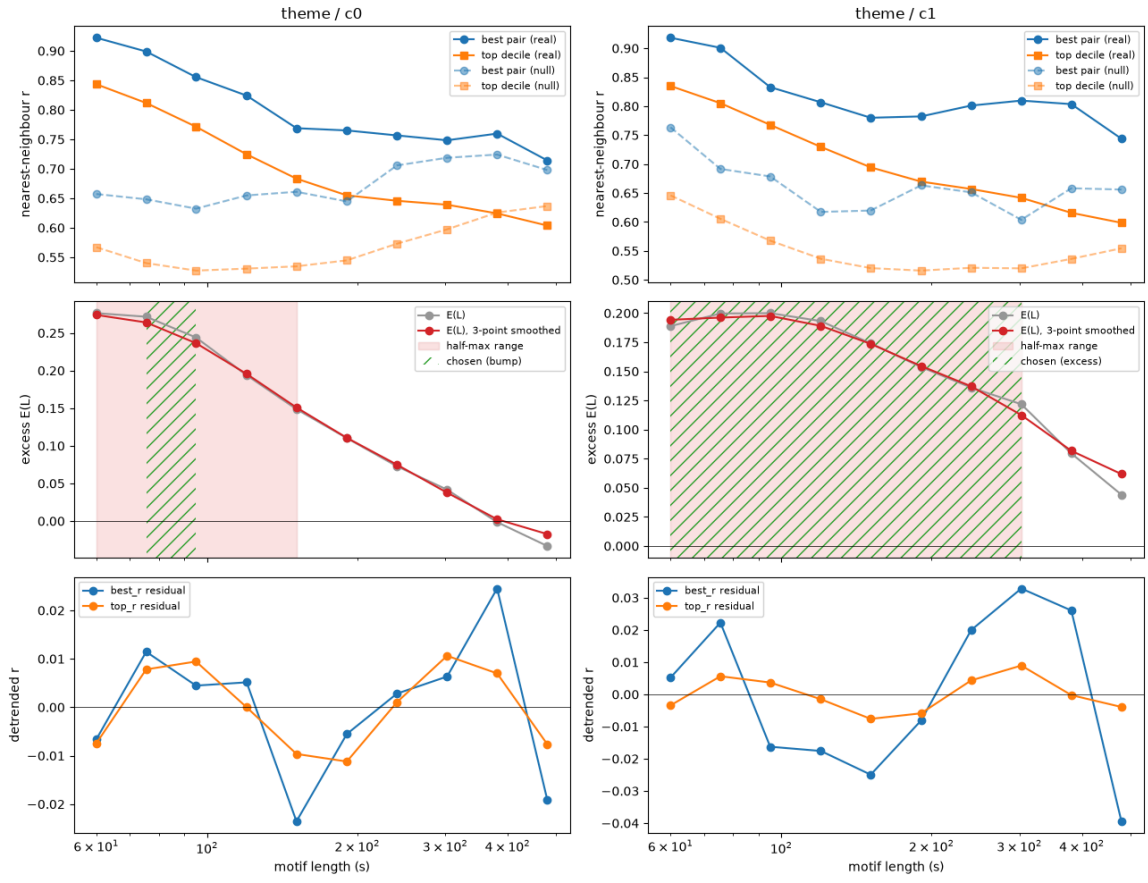
E(L) has a clear interior maximum: **0.28 at 38 s (c0)** and **0.19 at 50 s (c1)**. The half-maximum range covers 8.9–120 s for both, supported by correlation bumps (c0: 15.9, 50, 90 s; c1: 8.9, 21, 38, 50, 90 s) and by periods.

L (s)	5	8.9	15.9	28.3	37.8	50.4	67.3	120
E, c0	0.08	0.16	0.23	0.27	**0.28**	0.27	0.24	0.15
E, c1	0.07	0.10	0.14	0.16	0.18	**0.19**	0.16	0.15

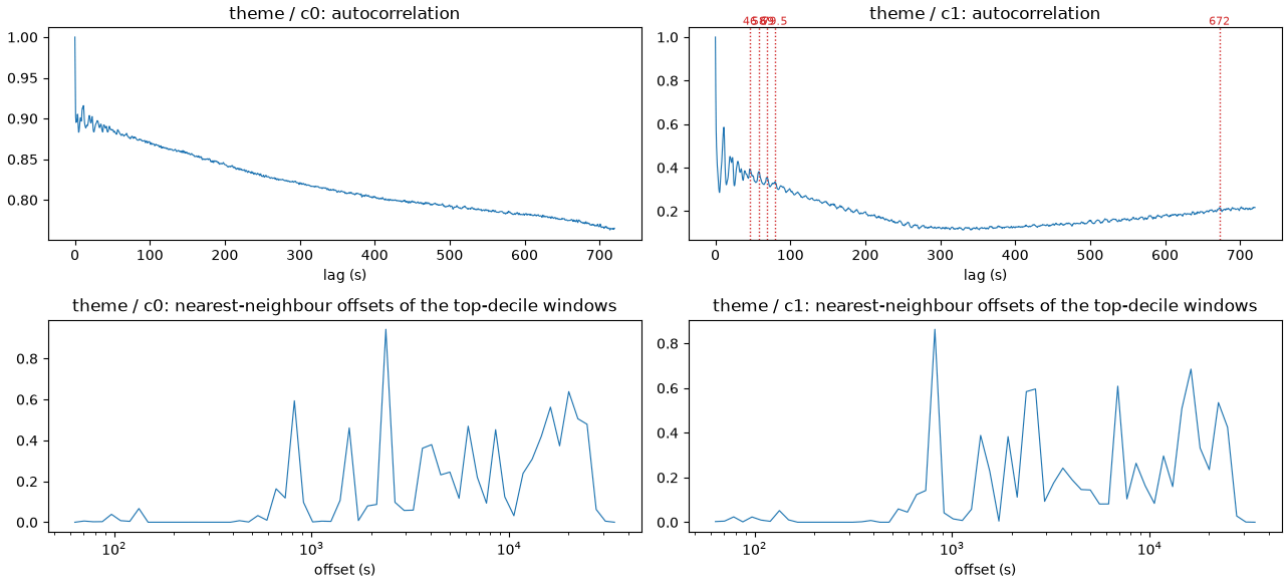


The autocorrelation has a comb of peaks at **~10–11 s and its multiples** (c0: 9.8, 19.3, 29.3, 38.2 s; c1: 11.1, 19.4, 29.8, 45.6, 57 s). This is the signature of phrases of ~10–11 s repeated several times in a row. The excess maximum at 40–50 s therefore covers about four repeated phrases.

Themes (1–8 min)

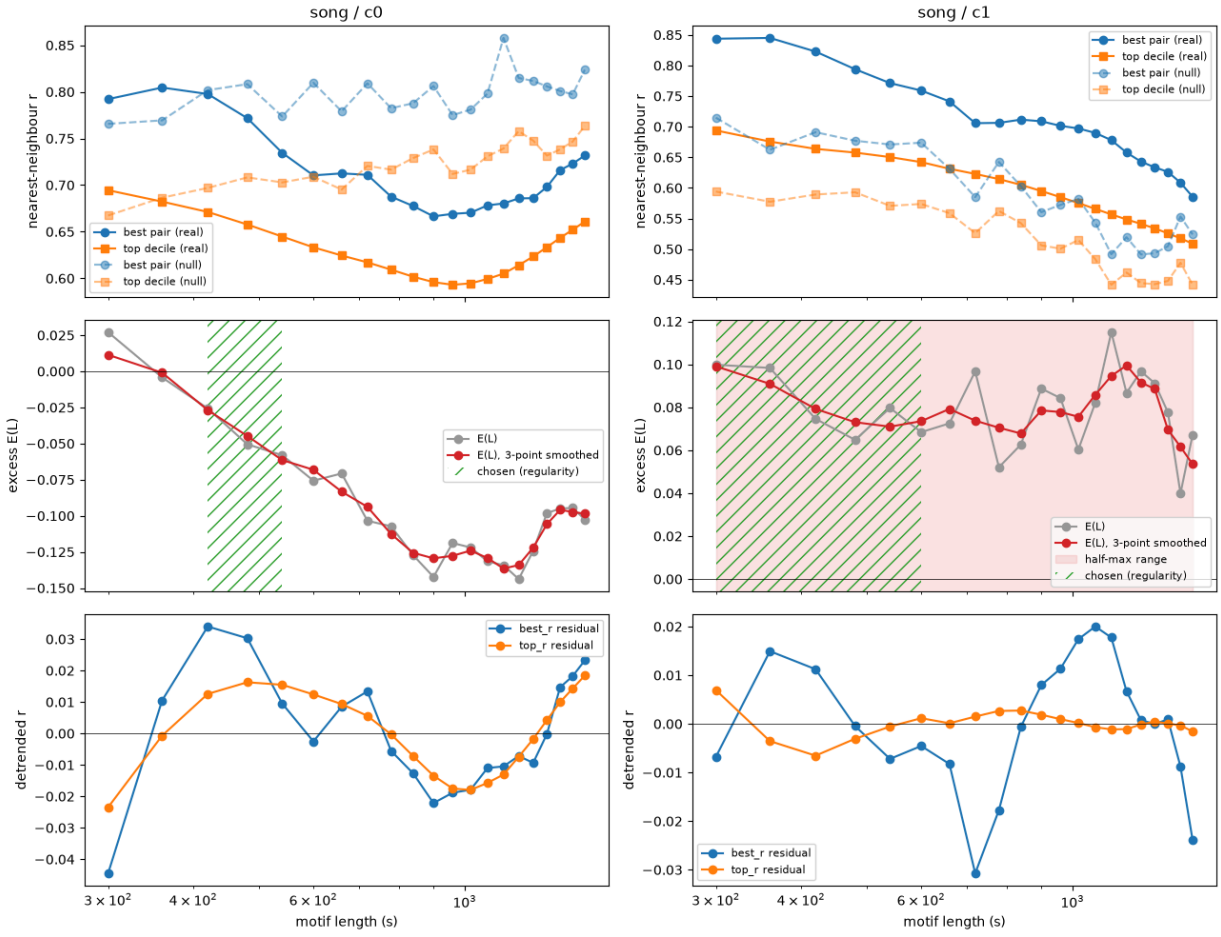


- **c0:** *E(L)* is largest at the shortest length (0.28 at 60 s) and falls to zero by ~6 min, i.e. it continues the phrase-band maximum. With no interior peak, the bump at 75–95 s decides.
- **c1:** a flat-topped maximum of 0.20 at 75–95 s, half-maximum range 60–300 s, supported by bumps at 75 and 300 s and by autocorrelation peaks at 46–80 s (still the ~11 s phrase comb).

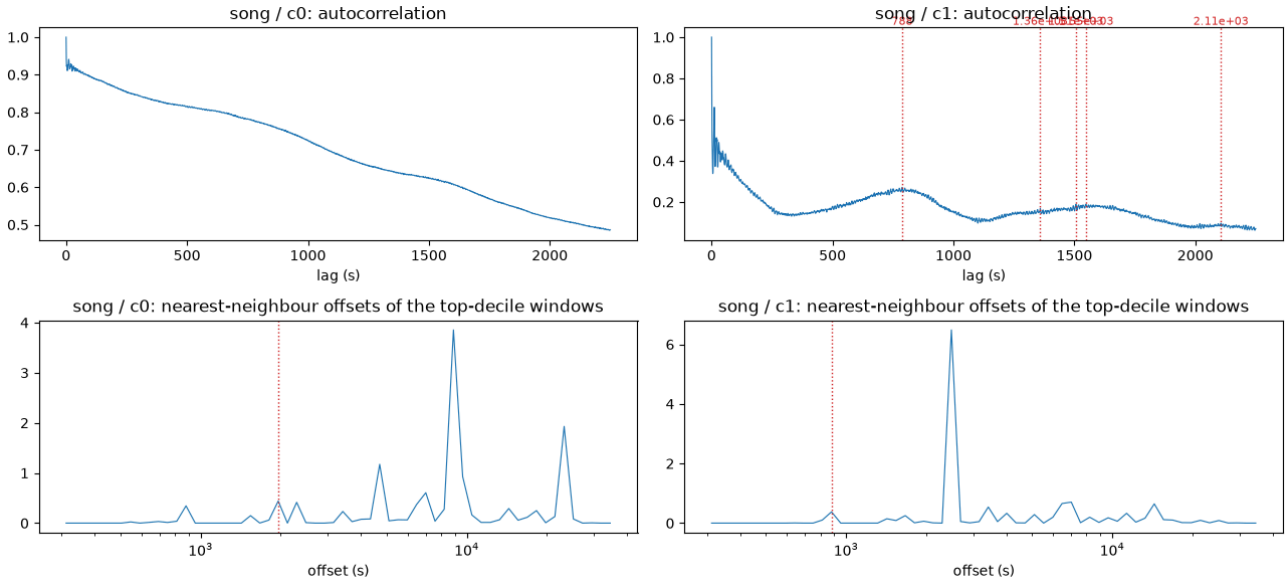


At this scale, c0's autocorrelation decays slowly from 0.9 (slow loudness changes). The offset histograms peak at **~820 s (13.6 min)** for both coefficients (c0's strongest peak is at ~2,400 s, about three cycles; the histogram bins are ~11% wide). The best match of a strongly repeating theme is usually the same theme in the next song.

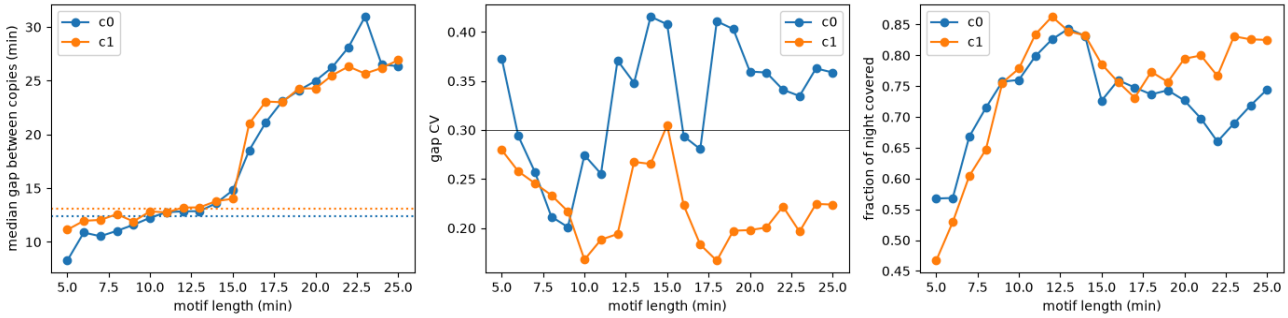
Songs (5–25 min)



- **c0:** the surrogate matches as well as or better than the data at every song length (*E* from +0.03 at 5 min down to −0.14). c0's slow loudness changes, which the surrogate keeps, dominate at this scale.
- **c1:** *E(L)* is positive but flat (0.04–0.12) across the band, so the half-maximum range spans everything and the excess criterion cannot pick a length.



c1's autocorrelation peaks at **788 s (13.1 min)** and again near 1,500 s (two cycles); c0 has no peaks. The regularity criterion therefore decides, using 13.1 min for c1 and the 12.4 min prior median for c0:



c1 motif length	5 min	6 min	7 min	8 min	9 min	10 min	11 min	12 min	16 min
copies	56	53	52	49	51	47	46	44	29
median gap (min)	11.1	12.0	12.1	12.6	11.9	12.8	12.8	13.2	21.0
gap CV	0.28	0.26	0.25	0.23	0.22	0.17	0.19	0.19	0.22
longest hole (min)	15.8	17.3	17.3	18.5	20.3	17.5	20.8	20.8	27.9
regular	yes	yes	yes	yes	no	yes	no	no	no

Up to ~15 min, copies recur every 11–14 min; above 15 min the gap jumps to ≥ 21 min (two cycles), because a motif longer than one song cycle can no longer find a copy in every cycle. The c1 regular lengths are 5–10 min (9 min narrowly fails on its 20.3 min hole). For c0, 7–9 min pass (gaps 10.5–11.6 min, CV 0.20–0.26); c0 at 10–15 min has very irregular gaps (CV 0.26–0.42) and holes of 30–51 min.

The chosen song motif lengths (5–10 min) are shorter than the full song (10–20 min). That is expected: a motif of about half a cycle is long enough to be unique to the song but short enough to be found in every cycle.

Caveats

- The excess criterion only resolves the phrase/theme scale. Units and songs are decided by the secondary criteria (bumps and regularity), which are noisier.

- The phrase range (8.9–120 s) is wide, because *E(L)* has a broad maximum. The Phase 2 search ranks lengths inside it by their excess, so it is not forced to the shortest length.
- Unit and phrase bands are scanned on segments (40 min and 3 h of the 10 h), so rare phrase types may be missed.
- The surrogate keeps the spectrum but not the non-stationarity (singing vs. silence). Part of the excess at long lengths may come from the on/off structure of singing bouts rather than from song content.

Reproducing

- Run the notebook from the `Whales_Song` folder. The first run takes ~20 minutes on a 32-core CPU (MFCC features ~1 min, unit scans ~9 min, the rest ~10 min). Results are cached (`whale_scan_<level>_<feature>.csv`, `whale_scan_<level>_<feature>_offsets.npz`, `whale_song_regularity_<feature>.csv`), so re-runs take ~15 s.
- Helpers live in `whale_mfcc.py`: `load_or_compute_features`, `find_offset`, `length_scan`, `phase_surrogate`, `detrended_residual`, `autocorrelation`, `period_peaks`, `offset_histogram`, `song_regularity`, `regular_range`, `choose_range`, `bump_range`.